

D Interpretability Analysis

This section provides the full methodology underlying the interpretability analyses summarized in Section 5. Specifically, we study: (i) per-layer attribution of the detector, (ii) logit-lens decoding of steering vectors, (iii) a hand-crafted stylistic-feature baseline, and (iv) token-level visualization of where the detection signal concentrates within a text. Unless stated otherwise, all analyses use the same frozen reference model (GPT-Neo-2.7B) and the same trained detectors as in the main experiments.

D.1 Per-layer contribution to the detector

Setup. Our detector is a logistic regression applied to layer-wise projection scores. For DetectRL, we use one feature per transformer layer:

$$\mathbf{s}(x) = (s_1(x), \dots, s_L(x)) \in \mathbb{R}^L,$$

$$s_\ell(x) = \frac{\langle a_\ell(x), v_\ell \rangle}{\|a_\ell(x)\|_2},$$

where $a_\ell(x)$ is the mean-pooled residual representation at layer ℓ , and v_ℓ is the corresponding steering vector. For MIRAGE, we retain three orthonormalized directions per layer (one per task), so $\mathbf{s}(x) \in \mathbb{R}^{L \times 3}$ is flattened before being passed to the classifier. In all cases, we train a standardized logistic regression (StandardScaler followed by ℓ_2 -regularized LogisticRegression with liblinear solver).

Attribution. After fitting the classifier, we attribute importance to each scalar feature using the magnitude-weighted coefficient

$$c_\ell = |w_\ell| \cdot \sigma_\ell,$$

where w_ℓ is the learned coefficient and σ_ℓ is the empirical standard deviation of feature ℓ on the training set. This quantity is invariant to feature rescaling and corresponds to the effective contribution of a standardized feature to the classifier logit. For MIRAGE, we sum c_ℓ across the three task-specific directions to obtain a single per-layer attribution score.

Fig. 9 shows that, on MIRAGE, the detector’s signal is concentrated in the final block of layers (L29-31), consistent with prior observations that higher-level stylistic features tend to emerge closer to the model output. On DetectRL, the contribution is more distributed: there is a sharp peak at L0-2,

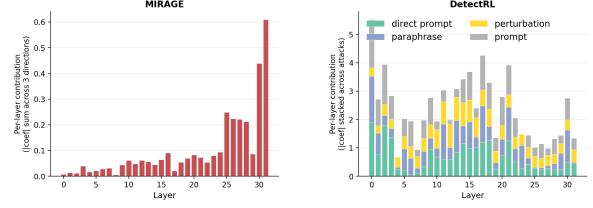


Figure 9

which we associate with surface-level lexical cues, a broader ridge at L14-18, and a secondary peak around L21. This profile motivates our choice of L25–31 for MIRAGE and L14-18 for DetectRL as the main targets of the logit-lens analysis below.

D.2 Logit-lens decoding of the steering vectors

Method. We apply a standard logit-lens probe (nostalgebraist, 2020) to each layer- ℓ steering vector v_ℓ by projecting it through the model’s final layer norm and unembedding head:

$$\text{logits}_\ell = W_U^\top \text{LN}_f(v_\ell) \in \mathbb{R}^{|\mathcal{V}|},$$

where $|\mathcal{V}| = 50,257$ is the GPT-Neo-2.7B vocabulary size. The interpretation is directional: the top-ranked indices correspond to tokens aligned with the $+v_\ell$ direction, which we interpret as *LLM-leaning*, while the bottom-ranked indices correspond to tokens aligned with $-v_\ell$, i.e. the *human-leaning* direction. We report raw logits rather than softmax probabilities, since the goal is to inspect relative alignment with the direction rather than recover a calibrated next-token distribution.

Results. Fig. 8a shows the top six tokens on each side. The resulting patterns are consistent across the selected layers:

- **MIRAGE, $-v_\ell$ (human-leaning).** Stylized closing punctuation (!", . . .", ').'), formal sentence-ending subwords (authorised, itially), and decorative or exaggerated tokens (£, ?????).
- **MIRAGE, $+v_\ell$ (LLM-leaning).** Surnames and journal abbreviations from the arXiv-derived split together with fragments more characteristic of academic prose (baugh, foreseen, paralleled, flavorful).
- **DetectRL, $+v_\ell$ (LLM-leaning).** Subword fragments of formal or emphatic vocabulary (astical, TRY, ermanent, ebted) and atypical punctuation or Unicode glyphs.

- **DetectRL, $-v_\ell$ (human-leaning).** More colloquial or topical fragments, including organising, Playstation, traged, adolesc, and bigot.

We stress that token-level logit-lens readings are noisy at any single layer. The patterns above should therefore be read as the consensus across the top contributing layers, rather than as literal interpretations of isolated tokens. Tokens that appear only at one layer are likely artifacts and should not be over-interpreted.

D.3 Stylistic-feature baseline

To quantify how much of the detector’s signal can be explained by hand-recognizable surface cues, we train a baseline logistic regression on a fixed bank of regex-derived features and compare its in-distribution AUROC to the full steering vector pipeline.

Feature bank. For each text, we compute counts and per-1,000-character rates for the following classes of patterns:

- **Typography:** em-dash usage (–), curly quotes, Unicode ellipsis (. . .) versus three-dot ellipsis (...), and space-before-period.
- **Markdown / formatting:** bold, italics, inline code, fenced code blocks, headers, and Oxford-comma lists.
- **Discourse:** formal connectives (Moreover, Furthermore, Additionally, In conclusion, Overall), summary prefixes, casual hedges (kinda, sorta, tbh), and inclusive pronouns (we, us, our).
- **Lexical:** “polished paraphrase” phrasing (e.g. utilizing, endeavor, showcase), consonant-cluster words, and log character length.
- **Prompt leakage and template artifacts:** [Assistant]: prefixes, templated review phrasing, generic summary formulas, and refusal-style stubs.
- **Mid-word case flips:** regex patterns that detect uppercase letters inside otherwise lower-case words, a known artifact under aggressive perturbation attacks.

We standardize the resulting feature vectors and fit an ℓ_2 -regularized logistic regression, tuning the regularization strength C by 5-fold cross-validation on a held-out split.

Results. Table 3 reports the in-distribution AUROC of this stylistic-feature baseline across all evaluated settings, together with the full steering vector pipeline for reference; the corresponding dumbbell visualization is shown in Fig. 8b. Two observations stand out:

1. Stylistic features alone form a non-trivial baseline, reaching 76–82 AUROC on MIRAGE and 88–91 AUROC on DetectRL. This is consistent with prior work showing that surface-level statistical and stylistic cues can already carry substantial signal for distinguishing machine-generated from human-written text.
2. The full steering vector pipeline still improves on this baseline by 13–24 AUROC points on MIRAGE and 9–12 points on DetectRL. We interpret this remaining gap as the part of the signal that is present in hidden representations but not recoverable from the regex bank. At the same time, the highest-weight regex features overlap substantially with the tokens surfaced by the logit-lens analysis (Fig. 8), including em-dashes, formal connectives, and prompt-leakage templates.

D.4 Cross-LM Logit-lens decoding of steering vectors

The main interpretability analyses in Sec D.1 decode steering vectors through the same reference LM that produced them, namely GPT-Neo-2.7B. A natural question is whether the human-vs-AI directions learned by SV-Detect are specific to GPT-Neo’s residual stream, or whether similar directions emerge in other modern LMs. To address this, we retrain the steering vector pipeline independently on several additional reference LMs and then decode the resulting dense steering vectors through each model’s own unembedding.

Reference LMs and datasets. We use four reference LMs spanning three families: EleutherAI/gpt-neo-2.7B, Qwen/Qwen3-1.7B, google/gemma-3-1b-pt, and google/gemma-3-1b-it. For each LM, we extract per-layer mean-pooled residual activations on the corresponding in-domain training corpora

+ v_ℓ LLM-leaning (top-pooled per LM)				- v_ℓ human-leaning (top-pooled per LM)			
Gemma-3-IT	Gemma-3-PT	Qwen-3	GPT-Neo	Gemma-3-IT	Gemma-3-PT	Qwen-3	GPT-Neo
captivating	<eos>	newfound	the	stupid	saddo	probably	nasa
leveraging	utilizing	advancements	both	had	anyway	analysed	invests
unwavering	esteemed	llen	behaviors	doesn't	kinda	really	bolst
utilization	impactful	firsthand	color	really	probably	anyway	organise
utilizing	confira	unfavorable	multiple	other	presumably	basically	's/aid
fostering	utilize	favorable	utilizing	anyway	actually	anyhow	lawy
multifaceted	upcoming	prioritize	upon	this	obviously	sic	anecd
prioritization	leveraging	utilization	all	basically	necessarily	analyse	licences
<unused2157>	mesmerizing	endeavors	global	but	programme	then	nostalg
flavorful	prioritizing	utilizing	future	etc	bloke	etc	athlet

Figure 10: Top-pooled tokens per reference LM, separately for the $+v_\ell$ (LLM-leaning, left) and $-v_\ell$ (human-leaning, right) sides. For each LM, we pool the top- K tokens across the last third of layers and across all analyzed DetectRL / MIRAGE settings, then list the most frequent normalized tokens. Despite differences in family and tokenizer, the positive side consistently surfaces a polished LLM-style register, while the negative side is more casual and discourse-like.

and fit the standard logreg steering vector construction described in Sec. 3. This yields one dense steering vector per layer for each (LM, dataset) pair. GPT-Neo is analyzed on four datasets (MIRAGE GENERATE/POLISH/REWRITE and DetectRL direct_prompt), while the other three LMs are analyzed on these four settings plus four additional DetectRL per-domain subsets (arxiv, xsum, writing_prompt, yelp_review), for a total of 28 (LM, dataset) pairs.

Method. For each (LM, dataset, layer), we decode the steering vector through the LM’s own final layer norm and unembedding using logit-lens probe as described in Sec. D.2.

Because single-layer readings are noisy and strongly affected by tokenizer idiosyncrasies, we aggregate within each LM before comparing across LMs. Specifically, for each LM we pool the top- $K_{\text{pool}} = 8$ tokens at every layer in the last third of the network depth, separately by sign, then take the union across the LM’s datasets. Layer ranges are L17–25 for Gemma-3, L18–27 for Qwen3, and L21–31 for GPT-Neo. Before computing cross-LM overlap, we normalize tokens by stripping whitespace, lowercasing, applying NFKC normalization, dropping control/format characters, and filtering out pure punctuation as well as tokens with less than 50% ASCII letters. This removes tokenizer-specific artifacts and makes the consensus comparison more meaningful.

Results. Fig. 10 shows the top pooled tokens for each LM separately, and Fig. 7 summarizes the cross-LM consensus. On the positive side, the four LMs converge on a clear LLM-polished register.

Three tokens (*endeavors*, *utilization*, and *utilizing*) appear in the top pool of all four LMs, and a broader cluster including *captivating*, *cybersecurity*, *leveraging*, *optimization*, *favorable*, and *unfavorable* appears in three of the four. On the negative side, the consensus is weaker but still interpretable: the most stable tokens are casual discourse markers such as *anyway*, *basically*, *etc*, *maybe*, *really*, and *very*. Here, the three smaller modern LMs agree more strongly with one another, while GPT-Neo more often surfaces named-entity fragments and topical content.

Overall, this analysis suggests that the lexical contrast captured by SV-Detect is not specific to GPT-Neo. When the steering vector pipeline is re-trained independently on different LM families and their dense directions are decoded through each model’s own unembedding, the same broad semantic register reappears: LLM-polished, formal, and abstract language on the positive side, and more casual, discourse-like language on the negative side.

D.5 Per-token visualization

The token-level figure in the main paper (Fig. 1) is constructed as follows. Given a text x and the best-AUROC layer ℓ^* identified by an AUROC-probe on the held-out set, we forward x through the frozen reference model and record the per-token residual vector

$$h_{\ell^*}(x_t) \in \mathbb{R}^d \quad \text{for } t \in \{1, \dots, T\}.$$

Each token is then assigned the dot product

$$\rho_t = \langle h_{\ell^*}(x_t), v_{\ell^*} \rangle$$

where v_{ℓ^*} is the steering vector at layer ℓ^* . Tokens are colored according to ρ_t , with the scale normalized to the 70th percentile of $|\rho|$ in the corpus so that most informative tokens reach full saturation. Red denotes positive projection ($+v$, LLM-leaning) and blue denotes negative projection ($-v$, human-leaning). The classification banner above each text corresponds to the full text-level prediction produced by the trained detector used in the main experiments.

E Results on the COLING-2025 MGT benchmark

For completeness, we additionally evaluate SV-Detect on the COLING-2025 Multilingual MGT benchmark, English split (Wang et al., 2025). Unlike DetectRL and MIRAGE, this benchmark pools